

# YUXUAN ZHANG

Combat Designer / Gameplay Programmer

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## EDUCATION

### UNIVERSITY OF SOUTHERN CALIFORNIA

M.S. Computer Science • Los Angeles, CA • Aug. 2025 – Dec. 2027

### HUAZHONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Eng., Biomedical Engineering • Wuhan, China • Aug. 2021 – Jun. 2025

## EXPERIENCE

### GAME COMBAT DESIGN INTERN • NETEASE LEIHUO • JUSTICE ONLINE MOBILE FEB. 2026 – AUG. 2026

- **Skill Combat Design:** Designed two skills for an upcoming version and completed the full skill-logic configuration; the content is now in player testing and final pre-release iteration.
- **Pipeline Coordination and Quality Polishing:** Coordinated animation, VFX, audio, and technical integration end to end; used internal and core-player testing to polish action feel and visual tension through frame-data and hit-stop tuning.
- **3C Experience and Balance:** Contributed to the new-version protagonist 3C reconstruction, planning mechanic optimizations and numerical adjustments for existing skills to improve competitive combat.

## PROJECTS

### RETURN TO AZEROTH: WARCRAFT FAN-MADE ARPG • UNREAL ENGINE 5, C++, BLUEPRINT • JAN. 2026 – PRESENT

- **Video Demo;** refactored Warcraft IP content into a Souls-like framework and authored design and technical documentation for the personal UE5 RPG project.
- Built responsive character 3C, dynamic combo attacks, parry, and multidirectional dodge through a Blueprint rapid-prototyping and C++ optimization pipeline.
- Used GAS for scalable ability activation, Attribute Sets, and Gameplay Effects; used Behavior Trees and State Trees for boss phases and enemy AI.

### SHIMMER: 2D PLATFORMER GAME • UNITY 2D, C# • OCT. 2025 – NOV. 2025

- **Video Demo;** designed and implemented core gameplay, level structure, boss encounters, and narrative flow in a 21-day cycle from concept to shipped build.
- Built all C# systems from scratch, including player control, layer switching, save/load, and animated UI; established workflows for in-game fixes in under 10 minutes.
- Released localized PC, WebGL, and Android builds on [Itch.io](#) and [TapTap](#), reaching 60+ first-week downloads.

### STEAM SIMULATOR: CLICKING TYCOON GAME • UNITY 2D, C# • SEP. 2025 – OCT. 2025

- Solely designed and developed a digital-game-marketplace loop with dynamic pricing and economy systems, delivering a complete game-jam demo in three days.
- Implemented progression, event-driven mechanics, and responsive UI; completed 10+ balance passes based on player feedback.
- Submitted to [Ludum Dare 58](#), earning an average player rating of 3.5/5.0 across evaluation categories.

### GAME ENGINE OPTIMIZATION • C++, PRIMEENGINE • AUG. 2025 – PRESENT

- Extended gameplay systems with adaptive NPC AI and reusable modular physics, including multi-collider setups, gravity, and collision checks.
- Implemented a custom particle system plus frustum and occlusion culling, improving stress-test frame rate by more than 50%.
- Added full-body, partial, and additive animation-layer blending to reduce visual pops and improve perceived smoothness.

## TOOLS

Unity

Unreal Engine 5

Codex

Visual Studio

Git

SVN

VS Code

Google Cloud Platform

Blender

## LANGUAGES

C / C++

C#

Python

Java

RISC-V

Verilog

MATLAB

HTML / CSS

## SKILLS

Combat Design

Gameplay Design

Gameplay Programming

Object-Oriented Programming

Debugging

Game Engines

Tabletop Design

## GAME INTEREST

|                          |        |
|--------------------------|--------|
| Steam                    | 1500+h |
| World of Warcraft        | 1000+h |
| Elden Ring               | 200+h  |
| Overwatch                | 200+h  |
| Hearthstone Battleground | 150+h  |
| Hollow Knight            | 100+h  |
| Diablo IV                | 100+h  |

And many more...